Vaibhav Hariram

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EDUCATION

University of California, Berkeley — **Berkeley, CA**
B.A. Computer Science & Geography, Minor in Data Science — *Expected May 2029*
Relevant Coursework: COMPSCI 61A, COMPSCI 61B, COMPSCI 70, COMPSCI 168, MATH 54, DATA C8

EXPERIENCE

Intel — **Santa Clara, CA**
AI/ML Engineering Intern — *June 2026 – Present*
- Shipped deterministic eval platform for pgvector hybrid search: 59-query benchmark, 9 metrics, SHA-256 binding, merged
- Scaled eval 5→59 audited cases; 5× recall overstated (1.00→0.20); localized misses to SQL prefilter boundary via tracing
- Built ACL-safe semantic backfill for under-filled hybrid retrieval: nextafter strict sub-floor scoring, file-level dedup
- Shipped content-negotiated progressive Markdown responses; built permission-filtered ripgrep API over 18K-doc corpus

Berkeley Artificial Intelligence Research (BAIR) — **Berkeley, CA**
AI Researcher (Gorilla Project, RISE Lab) — *February 2026 – July 2026*
- Contributor to BFCL: function-calling benchmark cited by Anthropic, OpenAI, NVIDIA teams across 40+ papers
- Drafted BFCL V5 multimodal extension for agentic behavior: 6 task categories across audio, vision, 3-turn dialogue
- Designed 9 perception-to-tool-call use case failure tests across audio/vision: homophones, slot hallucination, EXIF shortcuts

Paprika — **Bay Area, CA**
Systems Engineer — *March 2026 – June 2026*
- Architected execution control layer for agents: deterministic trace capture, enforcement, and step-level replay
- Built replay diffing engine detecting divergence across 1K+ agent traces with sub-second mismatch attribution
- Deployed system to 25+ B2B engineering teams; processing 50+ autonomous agent runs daily in dev environments

Flywheel OS — **Greater London, UK**
Founding Engineer (remote) — *January 2026 – April 2026*
- Built MCP server platform exposing 4 context tools to Claude Desktop/Code with API-key, flags, and usage logging
- Wired voice and context through 3 backend services, eliminating generic drafting across 100% of SDK content

Railinc Corp. — **Raleigh-Durham, NC**
Software Engineering Intern (x2) — *Summer 2024 & 2025*
- Shipped full-stack features (React, Java, Spring Boot) for RIGIS platform serving 20+ clients; 99.8% uptime SLA
- Built ETL pipeline (Python, ArcGIS Pro) processing 100K+ stations across the US; reduced data cleaning by 75%

OPEN SOURCE

- **vLLM** — Proposed optimizations to the logprobs/prefix-matching paths; closed after maintainer review (PR #37227)
- **inspect_evals** (UK AISI) — Fixed judge failures scored as zeros in PaperBench rubrics; 0.480→0.600 (PR #2335)
- **NVIDIA/garak** — Fixed detector misalignment silently dropping scan attempts from reports (PR #2140)
- **geofilter** (author) — Numba JIT point-in-polygon engine: 10M points in 0.58s (17.7M pts/sec avg over 10 runs)

PROJECTS

RouteX | C++, PostGIS, PostgreSQL, XGBoost
- Implemented A*, Dijkstra, and CH routing in C++ over 2.1M+ OSM Bay Area road segments via pybind11
- Preprocessing generates 30K+ shortcuts; bidirectional queries resolve in 0.013ms avg vs 0.764ms (59x speedup)
- Built PostGIS spatial query layer for network extraction, spatial joins, and geofencing over 4.1M+ road geometries
- Trained XGBoost ETA model across 14 road classification types; validated on 839K segments with 13.24% MAPE

ImmigrationBench | Python, PostgreSQL, Evaluation Harnesses
- Built LLM eval for F-1/H-1B deadlines: 2,400 cases, deterministic 80-rule ground truth, primary-source verified
- Frontier models reach 61–78% exact-date accuracy; 23% of wrong answers cite nonexistent regulatory sections
- Built six-class error taxonomy for arithmetic, rule-selection, and citation-hallucination failures; released dataset + harness

NYC-Access | Python, OpenStreetMap, scikit-learn, deck.gl
- Built partitioned PostGIS pipeline assigning 1.1B TLC stops to 780K OSM footprints at 1M rows/sec
- Built 1,000-building Street View ground truth; calibrated difficulty model improved accuracy +19pp (ECE 0.03)
- Active-learning loop reached 90% entrance accuracy with 4x fewer operator labels than random sampling
- Cross-city transfer to Chicago showed a 14pp accuracy drop, quantifying geographic overfitting across unseen urban layouts

SKILLS & TOOLS

- **Languages & Frameworks**: Python, TypeScript, SQL, Java, C++, FastAPI, React, Next.js, Node.js, Spring Boot
- **AI & ML**: evaluation harnesses, benchmark design, RAG, MCP, scikit-learn, XGBoost
- **Infrastructure**: PostgreSQL, pgvector, PostGIS, Docker, Redis, Supabase, Vercel, OpenTelemetry, Linux, CI/CD
- **Interests**: Chelsea F.C., K-Pop, Anime, GeoGuessr, Cats, Sketching, Home Decor, Online Shopping, Travel, Quiz Bowl